

**SIMATS ENGINEERING**

**ASSESSMENT TOOL 3**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1704

**COURSE OUTCOME COVERED**

**CO2:** *Perform dry run testing to trace the execution of algorithms, analyze intermediate steps, and verify the correctness of processes. (BL3)*

*Assessment Tool Weightage: 40%*

Duration : 45 Minutes

**QUESTIONS:** 2

**Total Marks:**20

Annexure A

**DRY RUN TEST**

***Code of Conduct:***

*I **MITHRA DEVI S - 192524203** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

***S. Mithra Devi***

- Current node
- Open List
- Closed List
- $g(n)$

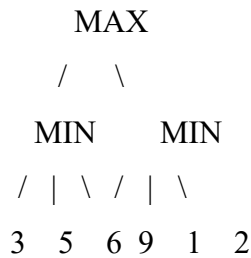
- $h(n)$
- $f(n) = g(n) + h(n)$

Finally,

1. Identify the optimal path.
2. Calculate the total path cost.

### Question 2: Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

Consider the following game tree.



#### Task:

Perform a **dry run of the Minimax algorithm with Alpha-Beta Pruning** by evaluating the tree **from left to right**.

For each node, show:

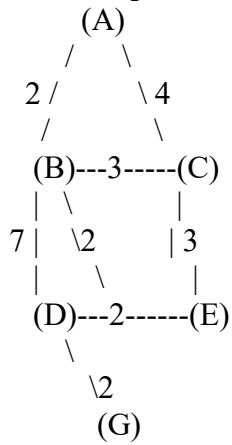
- Alpha ( $\alpha$ )
- Beta ( $\beta$ )
- Selected value
- Any node that gets pruned

Finally,

1. Determine the best move for MAX.
2. State the final minimax value.
3. List the pruned nodes (if any).

## Question 1: Dry Run of A Search Algorithm

### Given Graph



### Heuristic Values

Node  $h(n)$

|   |   |
|---|---|
| A | 7 |
| B | 6 |
| C | 4 |
| D | 3 |
| E | 2 |
| G | 0 |

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### Step 1

Current Node = A

- $g(A) = 0$
- $h(A) = 7$
- $f(A) = 0 + 7 = 7$

Expand A.

Children:

- $B \rightarrow g = 2$
- $C \rightarrow g = 4$

Node  $g(n)$   $h(n)$   $f(n)$

B    2    6    8

C    4    4    8

**Open List**

B(8), C(8)

**Closed List**

A

---

**Step 2**

Choose **B** (tie broken left to right)

Expand B.

New Nodes:

D

$$g(D)=2+7=9$$

$$f(D)=9+3=12$$

E

$$g(E)=2+2=4$$

$$f(E)=4+2=6$$

C already has cost 4

New cost through B

$$2+3=5$$

Not better

Ignore.

**Node g h f**

E 4 2 6

C 4 4 8

D 9 3 12

**Open List**

E(6), C(8), D(12)

**Closed List**

A, B

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### Step 3

Choose **E**

Expand E.

Possible paths

C

$$g=4+3=7$$

Current cost=4

Ignore

D

$$g=4+2=6$$

Current cost=9

Update D

G not directly connected.

### Node g h f

C 4 4 8

D 6 3 9

### Open List

C(8), D(9)

### Closed List

A, B, E

---

### Step 4

Choose **C**

Expand C.

No better path found.

### Open List

D(9)

### Closed List

A, B, E, C

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### Step 5

Choose **D**

Expand D.

Reach G

$$g(G)=6+2=8$$

$$f(G)=8+0=8$$

**Node g h f**

G     8 0 8

**Open List**

G(8)

**Closed List**

A, B, E, C, D

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### Step 6

Choose **G**

Goal reached.

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### Dry Run Table

| Iteration | Current Node | Open List         | Closed List |
|-----------|--------------|-------------------|-------------|
| 1         | A            | B(8), C(8)        | A           |
| 2         | B            | E(6), C(8), D(12) | A,B         |
| 3         | E            | C(8), D(9)        | A,B,E       |
| 4         | C            | D(9)              | A,B,E,C     |
| 5         | D            | G(8)              | A,B,E,C,D   |
| 6         | G            | Goal              | A,B,E,C,D,G |

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### Optimal Path

**A → B → E → D → G**

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**Total Path Cost**

$$A \rightarrow B = 2$$

$$B \rightarrow E = 2$$

$$E \rightarrow D = 2$$

$$D \rightarrow G = 2$$

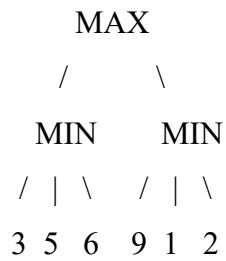
**Total Cost = 8**

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## Question 2: Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

### Game Tree



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### Initial Values

$$\alpha = -\infty$$

$$\beta = +\infty$$

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### Evaluate Left MIN

Leaf values

3

5

6

MIN chooses

$$\min(3,5,6)=3$$

Return

$$\text{Value} = 3$$

MAX updates

$$\alpha = \max(-\infty, 3)$$

$$\alpha = 3$$

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### Evaluate Right MIN

Current

$$\alpha = 3$$

$$\beta = +\infty$$

First leaf

9

$$\beta = \min(+\infty, 9) = 9$$

Since

$$\alpha = 3$$

$$\beta = 9$$

Continue

---

Second leaf

1

$$\beta = \min(9, 1) = 1$$

Now

$$\alpha = 3$$

$$\beta = 1$$

Since

$$\alpha \geq \beta$$

$$3 \geq 1$$

Prune remaining node

**2**

Return value

1

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## MAX Decision

Left child

3

Right child

1

MAX chooses

$$\max(3, 1) = 3$$

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### Dry Run Table

| Step | Node      | $\alpha$  | $\beta$   | Selected Value | Pruned    |
|------|-----------|-----------|-----------|----------------|-----------|
| 1    | MAX       | $-\infty$ | $+\infty$ | -              | -         |
| 2    | Left MIN  | $-\infty$ | 3         | 3              | None      |
| 3    | MAX       | 3         | $+\infty$ | 3              | None      |
| 4    | Right MIN | 3         | 9         | 9              | None      |
| 5    | Right MIN | 3         | 1         | 1              | Node 2    |
| 6    | MAX       | 3         | $+\infty$ | 3              | Completed |

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### Best Move for MAX

Choose the **Left MIN** subtree.

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### Final Minimax Value

**3**

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### Pruned Node

**Node 2**